Scalable Data Pipeline for Financial Risk Assessment using BigQuery & Power BI

Introduction

This report analyzes a dataset of 1 million+ loans to identify factors that lead to defaults. The analysis was performed using SQL for data preparation and Power BI for visualization.

ASK

Credit Portfolio Analysis - Business Questions

1. What is the Non-Performing Loan (NPL) Ratio?

- **Question:** What percentage of total disbursed loans are currently past due or in default ("bad debt")?
- **Why it matters:** This is the primary indicator of a bank's financial health and the quality of its credit assets.

2. How does the Credit Score (or Grade) impact the Probability of Default?

- **Question:** How much more frequently do customers with lower ratings (e.g., Grade E, F, G) stop paying compared to Class A customers?
- **Why it matters:** It validates whether the bank's internal risk assessment system (Credit Scoring) is accurate and reliable.

3. What is the relationship between Debt and Income (DTI)?

- **Question:** Is there a specific DTI threshold beyond which customers become excessively risky?
- **Why it matters:** This helps the bank establish strict regulatory limits and internal policies for the approval of new loans.

4. What is the purpose of the highest-risk loans?

- **Question:** Are "debt consolidation" loans riskier than "home improvement" loans?
- **Why it matters:** This data supports marketing strategies and risk-based pricing (setting interest rates based on the loan's purpose).

5. What is the geographic distribution of risk?

- **Question:** Are there specific geographic areas (states or cities) that show systematically higher insolvency rates?
- **Why it matters:** Essential for strategic portfolio management and for identifying regional economic crises or downturns.

Banking Glossary for Data Projects (Cheat Sheet)

1. Basic Loan Terms

- **Principal:** The original amount of money borrowed by the customer.
- **Interest Rate:** The percentage charged on the principal by the bank as the cost of borrowing.
- **Installment:** The fixed amount the customer pays every month.
- **Term:** The number of months scheduled for repayment (e.g., 36 or 60 months).

2. Loan Status

- **Fully Paid:** The loan has been repaid in full, including interest.
- **Current:** The customer is paying regularly and on schedule.
- **Late / In Grace Period:** A delay in payment (usually categorized between 16 and 120 days).
- **Default / Charged-off:** The loan is considered permanently unpaid; the bank has written it off as a loss.

3. Risk Indicators and Valuation

- **Loan Grade:** A bank's internal categorization of risk (e.g., A = low risk, G = high risk).
- **Debt-to-Income Ratio (DTI):** The percentage of a borrower's monthly income that goes toward paying debts. A higher DTI indicates higher risk.
- **Annual Income:** The yearly income reported by the borrower.
- **Non-Performing Loan (NPL):** Loans where collection is uncertain, typically after being overdue for more than 90 days.

4. Loan Purpose

- **Debt Consolidation:** Combining multiple high-interest debts into a single loan with a single monthly payment.
- **Credit Card:** Using the loan to pay off existing credit card balances.
- **Home Improvement:** Funding maintenance, repairs, or upgrades to a residence.

I DOWNLOADED THE CSV FILE FROM KAGGLE. I extracted it so it could be smaller andreadable in big query. Then i storaged on google cloud console and i created a bucket.

Data Cleaning & Preparation

- **Data Selection:** choose of the right columns (id, loan_amnt, int_rate, grade, addr_state, κτλ).
- **The `is_npl` Logic:**
  - *Definition:* Loans marked as "Charged Off" or "Late" were flagged as 1 (NPL), while "Fully Paid" and "Current" were flagged as 0.
- **Date Formatting:** changed the column `issue_d` to date

Create a binary flag using sql logic by organising npl as default loans, charged off or 30 or more days delay.

1. i cleaned the data with what is only needed.

```
SELECT
  id,
  loan_amnt,
  -- 36 mesi a 36
  SAFE_CAST(REGEXP_EXTRACT(term, r'\d+') AS INT64) AS term_months,
  int_rate,
  installment,
  grade,
  emp_length,
  home_ownership,
  annual_inc,
  verification_status,
  issue_d,
  loan_status,
  purpose,
  addr_state,
  dti,
  -- rossi mutui (NPL)
  CASE
```

```
    WHEN loan_status IN ('Charged Off', 'Default', 'Late (31-120 days)', 'Does not
meet the credit policy. Status:Charged Off') THEN 1
    ELSE 0
  END AS is_npl
FROM
  `civil-icon-433816-u2.banking_data.final_loan_analysis`
WHERE
  id IS NOT NULL
```

Then I made some practice on sql but it will not have an impact on the results.
1:Purpose of npl ratio (Purpose);

```
SELECT
  purpose,
  COUNT(id) AS total_loans,
  AVG(is_npl) AS npl_ratio
FROM
  `civil-icon-433816-u2.banking_data.final_loan_analysis`
GROUP BY
  purpose
ORDER BY
  npl_ratio DESC;
```

2: Average credit score and interest?

```
SELECT
  grade,
  AVG(int_rate) AS avg_interest_rate,
  AVG(is_npl) AS risk_level
FROM
  `civil-icon-433816-u2.banking_data.final_loan_analysis`
GROUP BY
  grade
ORDER BY
  grade ASC;
```

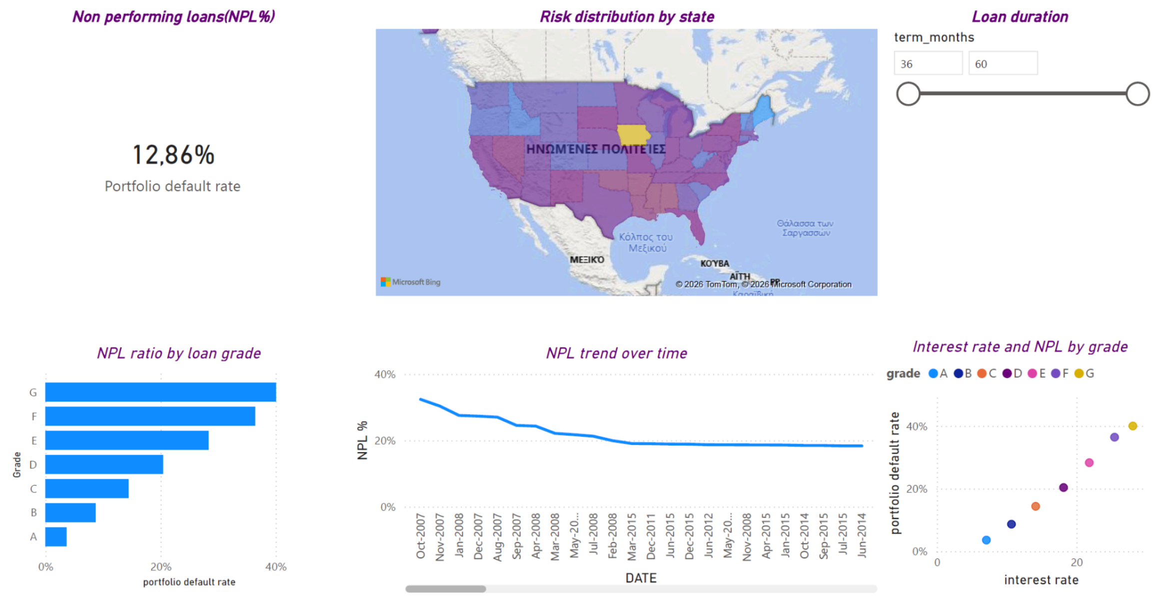
3: (String Manipulation)

```
SELECT
  term,
  SAFE_CAST(REGEXP_EXTRACT(term, r'\d+') AS INT64) AS term_months
FROM
  `civil-icon-433816-u2.banking_data.final_loan_analysis`
LIMIT 10;
```

Executive Summary

- **Total NPL Ratio:** 12.86%.
- **Average Interest Rate:** (Βάλε το μέσο επιτόκιο που σου βγάζει η κάρτα σου στο Power BI).
- **Loan Volume:** Πόσα δάνεια αναλύθηκαν συνολικά.

Key Visualizations & Analysis



- **Risk by Grade:** "As expected, lower grades (E, F, G) show a much higher NPL ratio, reaching up to X%."
- **Interest Rate vs NPL:** "There is a clear positive correlation. The bank charges more for higher risk, which is a correct pricing strategy."
- **Geographic Insights:** "States like California and Texas show the highest volume, while [βάλε μια πολιτεία με υψηλό ρίσκο] shows the highest default rate."

Conclusion & Recommendations

- Review the 60-month loan product due to higher default rates.
- Implement stricter criteria for Grade G loans.